Homework #4

Problem1

In this exercise, we examine how pipelining affects the clock cycle time of the processor. Problems in this exercise assume that individual stages of the datapath have the following latencies:

IF	ID	EX	MEM	WB
250ps	400ps	150ps	350ps	200ps

Also, assume that instructions executed by the processor are broken down as follows:

alu	beq	lw	sw
40%	20%	25%	15%

a) What is the clock cycle time in a pipelined and non-pipelined processor?
b) What is the total latency of an LW instruction in a pipelined and non-pipelined processor?
c) If we can split one stage of the pipelined datapath into two new stages, each with half the latency of the original stage, which stage would you split and what is the new clock cycle time of the processor?

解: (a) pipelined: 400ps

non-pipelined: 250ps+400ps+150ps+350ps+200ps = 1350ps

(b) pipelined: 400ps×5 = 2000 ps

non-pipelined: 1350ps

(c) ID, 350ps

Problem 2

Look for data hazards in the code below, and figure out how forwarding could be used to solve them.
them.

Instruction	C1	C2	C3	C4	C5	C6	C7
1. addi t0, a0, -1	IF	ID	EX	MEM	WB		
2. and s2, t0, a0		IF	ID	EX	MEM	WB	
3. sltiu a0, t0, 5			IF	ID	EX	MEM	WB

Data Hazard:

t0 a0 -1
s2 t0 a0
a0 t0 5

① Forwarding the EX in C3 to EX in C4

② Forwarding the MEM in C4 to EX in C5

Problem 3

This exercise is intended to help you understand the relationship between forwarding, hazard detection, and ISA design. Problems in this exercise refer to the following sequence of instructions, and assume that it is executed on a 5-stage pipelined datapath:

```
add x5, x2, x1
lw x3, 4(x5)
or x3, x5, x3
lw x2, 0(x2)
sw x3, 0(x5)
```

1) if there is no forwarding or hazard detection, insert NOP to ensure correct execution.
2) Repeat 1) but now use NOP only when a hazard cannot be avoided by changing or rearranging these instructions. You can assume register x7 can be used to hold temporary values in your modified cade.
3) If the processor has forwarding, but we forget to implement the hazard detection unit, what happens when this code executes?

解:

1)
```
add x5, x2, x1
nop x2
lw x3, 4(x5)
nop x2
or x3, x5, x3
lw x2, 0(x3)
nop
sw x3, 0(x5)
```

2)
```
add x5, x2, x1
lw x2, 0(x2)
nop
lw x3, 4(x5)
nop x2
or x3, x5, x3
nop x2
sw x3, 0(x5)
```

3)
```
add x5, x2, x1
lw x3, 4(x5)
Hazard
or x3, x5, x3
lw x2, 0(x3)
sw x3, 0(x5)
```

若无hazard detect or x3, x5, x3 中的 x3 还来被load就用作计算了

Problem 4

In the conventional fully bypassed 5-stage pipeline discussed in class, the main remaining data hazard is the load-use hazard. For example, in the following instruction sequence:

```
lw x1, 16(x2)
xor x3, x1, x5
```

The xor instruction will experience a one-cycle stall in the decode stage, while the lw propagates to the memory stage. For the remainder of this question, we will only consider 32-bit loads (lw). ***You may ignore branch and jump instructions in this problem.***

```
F   D   X   M   W                   lw
    F   D   D   X   M   W           xor
```

v. 猜测

One approach to removing the hazard is to speculate on the load value returned (*load-value speculation*). In lecture, we briefly described one scheme, a *load-zero predictor* where the value was predicted to be zero. An instruction that would otherwise stall in the decode stage waiting for a memory value was provided the value 0 instead. When the load instruction reaches the memory stage, the value is checked and if not zero, the pipeline is flushed and the dependent instruction is replayed as shown.

```
F   D   X   M   W                           lw
    F   D   X   -   -                       xor flushed
        F   D   -   -   -                   flushed
            F   -   -   -   -               flushed
                F   D   X   M   W           xor correct value
```

Q3.A [3 points] Assuming the pipeline is flushed as shown on a mispredict, calculate the minimum accuracy required for the load-zero predictor to improve performance. Note the predictor is only used when an instruction would otherwise stall in decode waiting for a load value.

1

解. $2r + 5(1-r) = 3 \Rightarrow r = \frac{2}{3}$

The minimum accuracy is $\frac{2}{3}$